Alex James Thoden

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Active Top Secret/SCI Clearance with CI Polygraph

SUMMARY

Data scientist with ∼4 years building predictive models and decision-support frameworks for federal programs; experienced with billion-record datasets, Monte Carlo analysis, and cloud-scale ETL. Strong at translating complex findings for executive stakeholders. Tools: Python, R, SQL, AWS.

EDUCATION

Georgia Institute of Technology **July 2026**
M.S. Computational Data Analytics

Virginia Tech **May 2022**
B.S. Materials Science and Engineering

SKILLS

Programming and Tools: Python (Polars, Pandas, NumPy), scikit-learn, SciPy, PyTorch, SQL (PostgreSQL, SQL Server), DuckDB, R (tidyverse, ggplot2, Shiny), Dash, Flask, Tableau, PowerBI

Web Development: JavaScript, TypeScript, React, Vite, D3, Tailwind CSS

Cloud & DevOps: AWS (S3, CloudTrail, EC2, Lambda), Git/GitLab, CI/CD, Docker, Podman

AI Tooling: Agentic workflows, system prompt engineering, vector embedding; integrated secure generative AI into workflows

Analytics & Modeling: Statistical modeling, Monte Carlo simulation, forecasting, regression/classification, vectorization, data visualization, ML model creation & evaluation, AI integration

Interpersonal: Translate complex technical concepts for non-technical audiences; strong collaborator and communicator.

Research & Learning: Quickly adopt new tools and technologies to improve team output.

EXPERIENCE

Senior Associate Modeling and Simulation Engineer, Peraton **Sep 2024 – Present**

- Technical lead for enterprise analytical framework (Authoritative Source of Truth) delivering 10-year capacity forecasts that inform $100M+ annual budget decisions.
- Release quarterly predictive models integrating billions of records across operational systems and cloud platforms; own ETL and validation at scale.
- Designed Monte Carlo simulations across 8+ modeling scenarios, identifying cost optimization opportunities of 34–54%.
- Built parametric models of volumetric usage to support data governance policy decisions.
- Developed a React "what-if" app to accelerate risk/readiness assessments and scenario exploration.
- Led forecasting models for customer scaling needs; automated data connections so stakeholders receive refreshed insights without manual steps.
- Integrated Jira with secure generative AI to auto-produce impact statements, improving reporting quality and eliminating manual overhead for 20+ engineers.
- Delivered rapid executive analyses (often <6 hours), including AoA showing ∼65% cost reduction potential over 6 years via policy optimization.

- Built PowerBI dashboards for OMB responses and an auto-refresh briefing interface tied to the data warehouse to support real-time decisions.

Associate Modeling and Simulation Engineer, Peraton **Aug 2022 – Sep 2024**

- Implemented and maintained CI/CD runners for cross-domain pipelines, enabling workflows in NGA's Maven AI/ML platform across protected networks.
- Supported Maven platform transition into the NSG through systems engineering and integration testing.
- Established the inaugural Maven Engineering Review Board to improve program-wide delivery consistency.
- Architected production ETL pipelines processing terabyte-scale datasets daily; implemented automated data validation (e.g., Patito) across billions of records.
- Processed historical enterprise library data (2000–present) to baseline 10-year capacity forecasts; optimized file size/count queries for faster studies.
- Created a transaction cost model showing no significant added cost when migrating data to colder storage tiers.
- Built tooling that exposes XML configuration as editable CSV, enabling non-SMEs to adjust parameters and output valid XML, removing bottlenecks.

PROJECTS

Wedding website: Designed and developed full-stack website with secure RSVP functionality and admin dashboard to track response statistics. Built reusable React components, implemented a lightweight API for data storage, and added visualizations for monitoring of attendance trends. Impact: **Built with: React, TypeScript, Tailwind CSS**

Seating Chart Optimization Algorithm: Created an optimization engine using simulated annealing and genetic algorithms to generate optimized seating arrangements based on user-defined guest relationships through interactive GUI.
Built with: Python, React, TypeScript, Tailwind CSS